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Abstract

This project analyzes job market trends using Power BI. The dataset was examined to identify job demand, salary patterns, top industries, leading companies, required skills, and educational requirements. Interactive dashboards were created to support data-driven decision-making.

Introduction

The job market is evolving rapidly with increasing demand for data-driven decision making. Organizations seek professionals with expertise in analytics, machine learning, cloud computing, and business intelligence. This project provides a detailed analysis of hiring trends and workforce requirements.

Objectives

- Identify states with the highest number of jobs.
 - Analyze salary trends across states.
 - Determine top industries and companies.
 - Identify most demanded job titles.
 - Analyze skill requirements.
 - Study the relationship between education and salary.
 - Develop an interactive Power BI dashboard.
-

Dataset Description

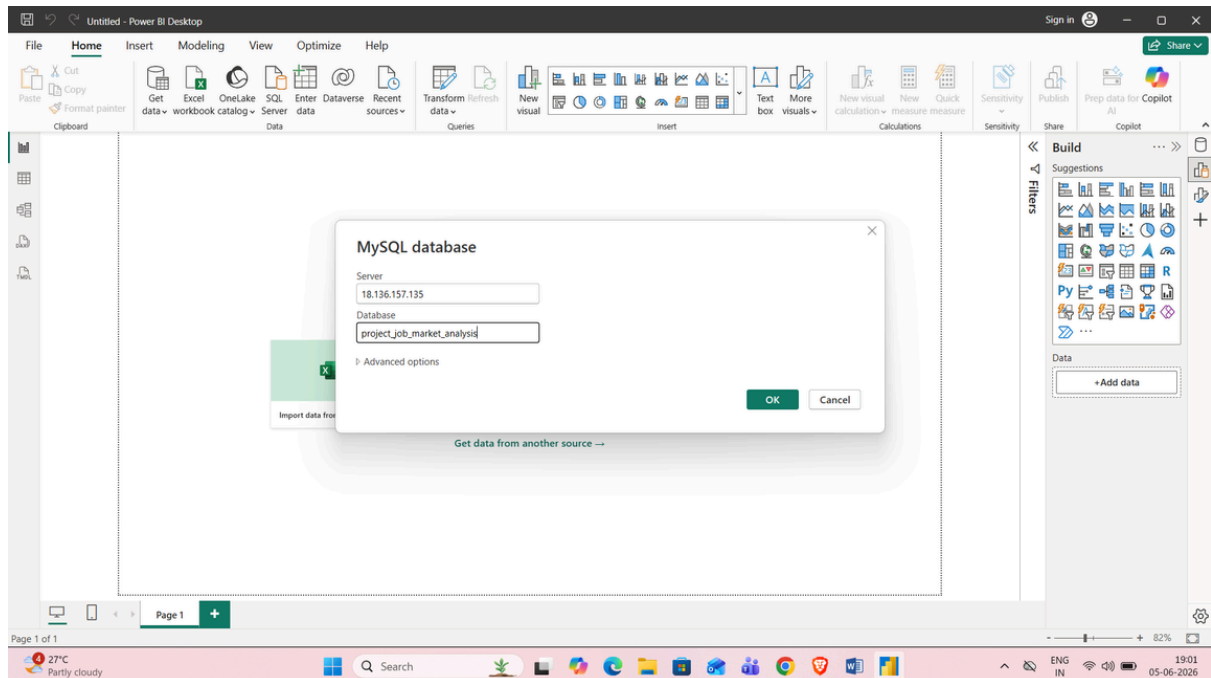
The dataset contains:

- Job Title
- Company Name
- Industry • Location
- Degree/Education
- Lower Salary
- Upper Salary
- Average Salary
- Skills such as SQL, Python, Spark, Tableau, TensorFlow, AWS, etc.

Data Cleaning and Preprocessing

The following steps were performed:

1. Removed duplicate records.
2. Handled missing values.
3. Corrected data types.
4. Validated salary fields.
5. Standardized categorical values.
6. Created calculated measures.
7. Prepared data for dashboard visualization.



Navigator

Display Options ▾

18.136.157.135: project_job_market_analysis [1]

project_job_market_analysis.Market

project_job_market_analysis.Market

ID	Job_Title	Salary_Estimate	FIELD4
0	Data Scientist	\$53K-\$91K (Glassdoor est.)	null
1	Healthcare Data Scientist	\$63K-\$112K (Glassdoor est.)	null
2	Data Scientist	\$80K-\$90K (Glassdoor est.)	null
3	Data Scientist	\$56K-\$97K (Glassdoor est.)	null
4	Data Scientist	\$86K-\$143K (Glassdoor est.)	null
5	Data Scientist	\$71K-\$119K (Glassdoor est.)	null
6	Data Scientist	\$54K-\$93K (Glassdoor est.)	null
7	Data Scientist	\$86K-\$142K (Glassdoor est.)	null
8	Research Scientist	\$38K-\$84K (Glassdoor est.)	null
9	Data Scientist	\$120K-\$160K (Glassdoor est.)	null
10	Data Scientist	\$126K-\$201K (Glassdoor est.)	null

The data in the preview has been truncated due to size limits.

Select Related Tables

Load

Transform Data

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File Home Help Table tools Column tools

Name ID Data type Whole number Format Whole number Count Summarization Count Data category Uncategorized

Structure Formatting Properties Sort by column Sort Data groups Manage relationships New column Calculations

ID	Job_Title	Salary_Estimate	FIELD4	Rating	Company_Name	Location	Headquarters	Size	Founded	Type of
0	Data Scientist	\$53K-\$91K (Glassdoor est.)		3.8	Tecolote Research 3.8	Albuquerque, NM	Goleta, CA	501 - 1000	1973	Com
1	Healthcare Data Scientist	\$63K-\$112K (Glassdoor est.)		3.4	University of Maryland Medical Syst	Linthicum, MD	Baltimore, MD	10000+	1984	Other
2	Data Scientist	\$80K-\$90K (Glassdoor est.)		4.8	Knowl4 4.8	Clearwater, FL	Clearwater, FL	501 - 1000	2010	Com
3	Data Scientist	\$56K-\$97K (Glassdoor est.)		3.8	PNNL 3.8	Richland, WA	Richland, WA	1001 - 5000	1965	Gove
4	Data Scientist	\$86K-\$143K (Glassdoor est.)		2.9	Affinity Solutions 2.9	New York, NY	New York, NY	51 - 200	1998	Com
5	Data Scientist	\$71K-\$119K (Glassdoor est.)		3.4	CyrusOne 3.4	Dallas, TX	Dallas, TX	201 - 500	2000	Com
6	Data Scientist	\$54K-\$93K (Glassdoor est.)		4.1	ClearOne Advantage 4.1	Baltimore, MD	Baltimore, MD	501 - 1000	2008	Com
7	Data Scientist	\$86K-\$142K (Glassdoor est.)		3.8	Logic20/20 3.8	San Jose, CA	Seattle, WA	201 - 500	2005	Com
8	Research Scientist	\$38K-\$84K (Glassdoor est.)		3.3	Rochester Regional Health 3.3	Rochester, NY	Rochester, NY	10000+	2014	Hosp
9	Data Scientist	\$120K-\$160K (Glassdoor est.)		4.6	<intent> 4.6	New York, NY	New York, NY	51 - 200	2009	Com
10	Data Scientist	\$126K-\$201K (Glassdoor est.)		3.5	Wish 3.5	San Jose, CA	San Francisco, CA	501 - 1000	2011	Com
11	Data Scientist	\$64K-\$106K (Glassdoor est.)		4.1	ManTech 4.1	Chantilly, VA	Herndon, VA	5001 - 10001	1968	Com
12	Staff Data Scientist - Technology	\$106K-\$172K (Glassdoor est.)		3.2	Walmart 3.2	Plano, TX	Bentonville, AR	10000+	1962	Com
13	Data Analyst	\$46K-\$85K (Glassdoor est.)		4.1	Yesler 4.1	Seattle, WA	Seattle, WA	201 - 500	2012	Com
14	Data Scientist	\$83K-\$144K (Glassdoor est.)		3.7	Takeda Pharmaceuticals 3.7	Cambridge, MA	OSAKA, Japan	10000+	1781	Com
15	Data Engineer I	\$102K-\$190K (Glassdoor est.)		3.6	Audible 3.6	Newark, NJ	Newark, NJ	1001 - 5000	1995	Subs
16	Scientist /IL Biology	\$67K-\$137K (Glassdoor est.)		3.9	Blueprint Medicines 3.9	Cambridge, MA	Cambridge, MA	51 - 200	2011	Com
17	Customer Data Scientist	\$118K-\$189K (Glassdoor est.)		4.3	h2o.ai 4.3	Mountain View, CA	Mountain View, CA	201 - 500	2011	Com
18	Data Scientist - Health Data Analytics	\$110K-\$175K (Glassdoor est.)		4.2	Nuna 4.2	San Francisco, CA	San Francisco, CA	51 - 200	2010	Com
19	Data Scientist	\$64K-\$111K (Glassdoor est.)		4	Pinnacle Assurance 4.0	Denver, CO	Denver, CO	501 - 1000	1915	Non
20	Data Scientist	\$81K-\$130K (Glassdoor est.)		3.2	Porch 3.2	Seattle, WA	Seattle, WA	201 - 500	2012	Com
21	Senior Data Scientist / Machine Learning	\$73K-\$119K (Glassdoor est.)		3.9	Health IQ 3.9	Dallas, TX	Mountain View, CA	201 - 500	2013	Com
22	Data Scientist - Quantitative	\$86K-\$139K (Glassdoor est.)		3.8	Truckstop.com 3.8	Chicago, IL	New Plymouth, ID	501 - 1000	1995	Com
23	Data Scientist	\$63K-\$105K (Glassdoor est.)		4.3	SMC 3 4.3	Louisville, KY	Peachtree City, GA	51 - 200	1935	Non
24	Data Scientist	\$109K-\$177K (Glassdoor est.)		4	Novetta 4.0	Herndon, VA	Mc Lean, VA	501 - 1000	2012	Com
26	Digital Health Data Scientist	\$63K-\$110K (Glassdoor est.)		4	Pfizer 4.0	Cambridge, MA	New York, NY	10000+	1849	Com
27	Data Scientist	\$75K-\$124K (Glassdoor est.)		3.5	First Tech Federal Credit Union 3.5	Hillsboro, OR	San Jose, CA	1001 - 5000	1952	Com

Table: project_job_market_analysis Market (742 rows) Column: ID (742 distinct values)

Search

project_job_market_analysis Ma...

Σ Age

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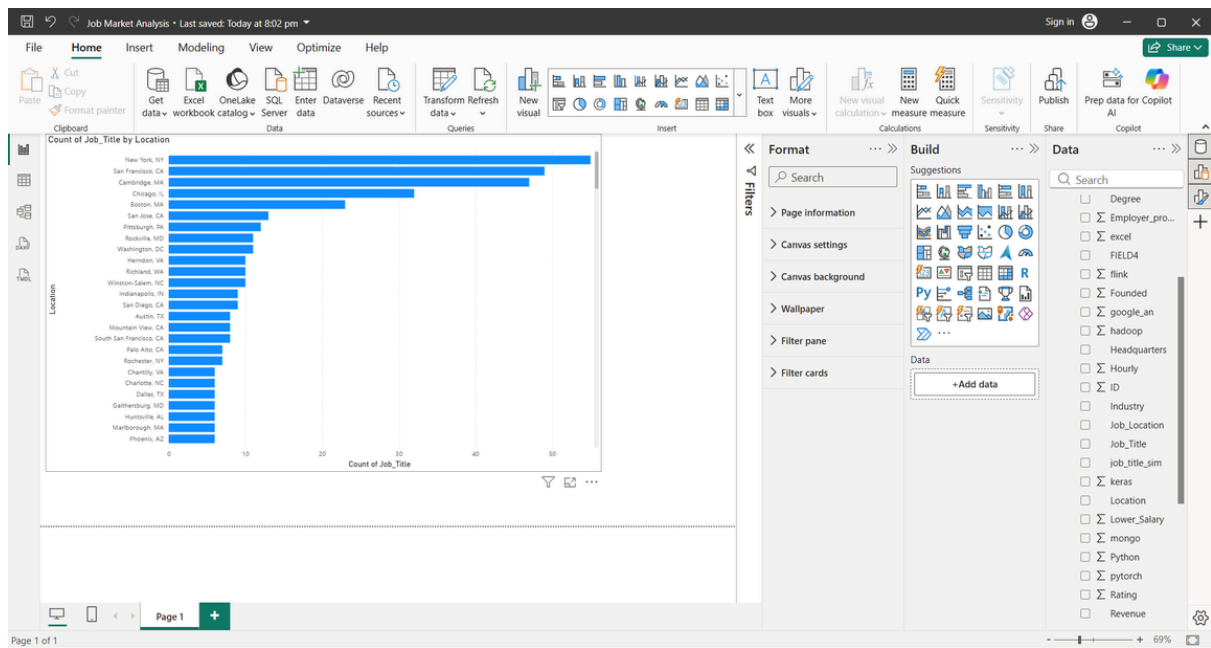
job title sim

Methodology

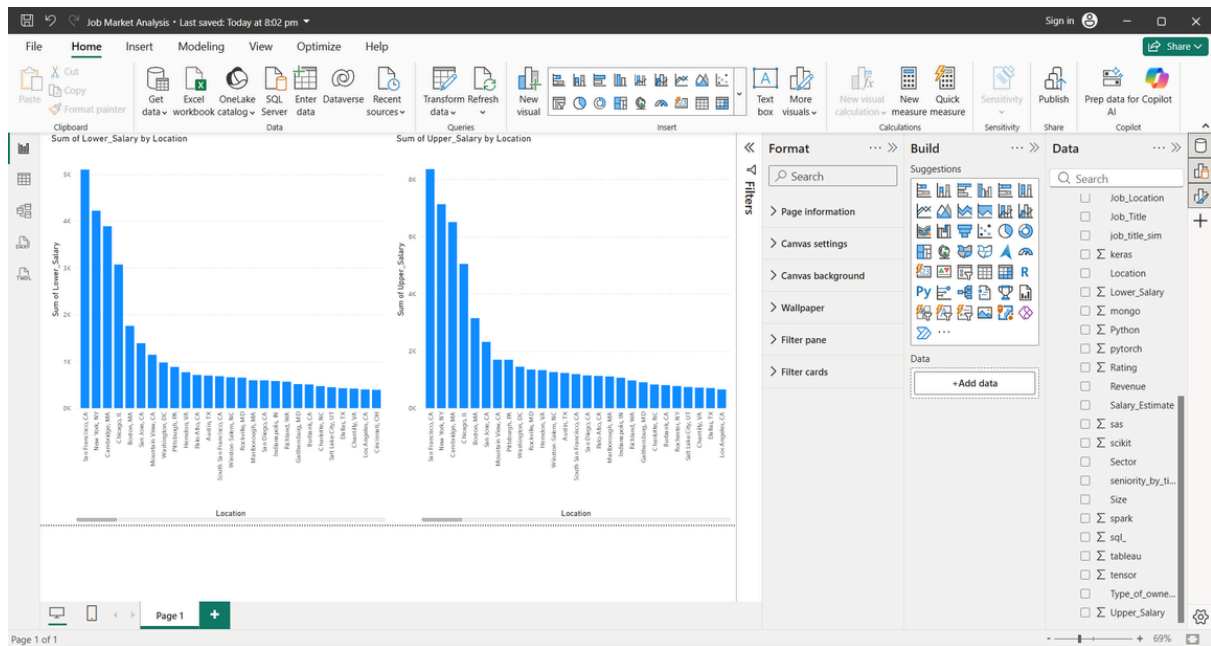
1. Data Import into Power BI.
 2. Data Cleaning using Power Query.
 3. Data Transformation.
 4. Creation of Measures and Calculated Columns.
 5. Visualization Development.
 6. Dashboard Design.
 7. Insight Generation and Recommendations.
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Analysis and Visualizations

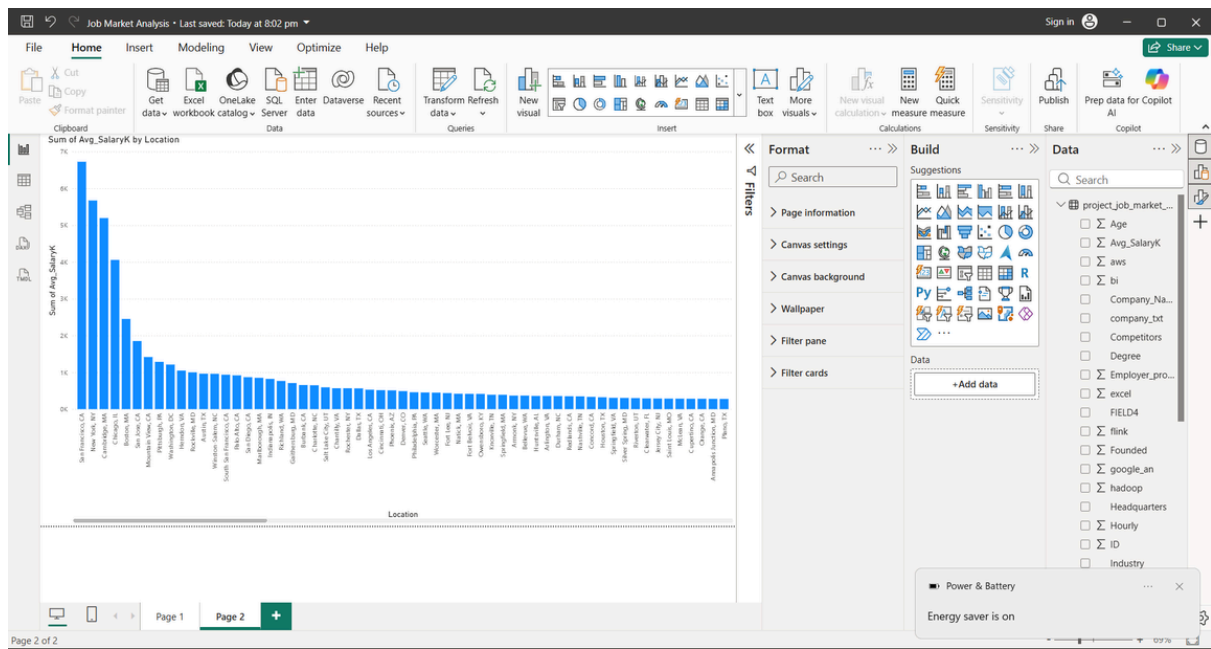
A. States with Most Jobs



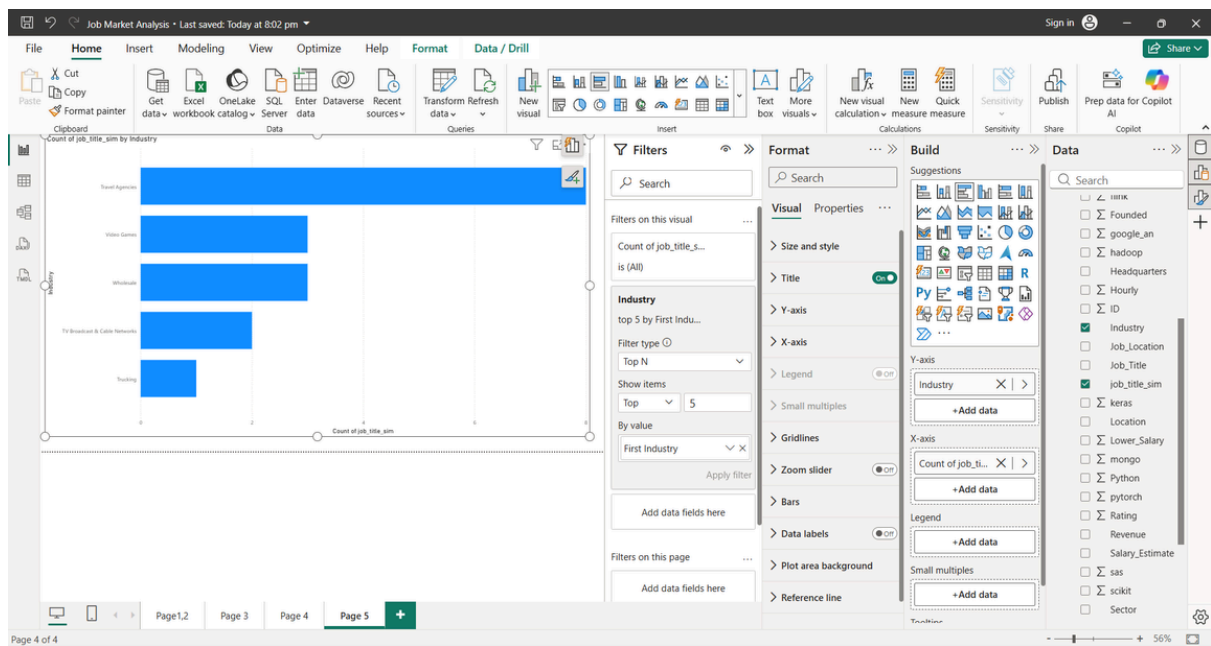
B. Average Minimum and Maximum Salary by State



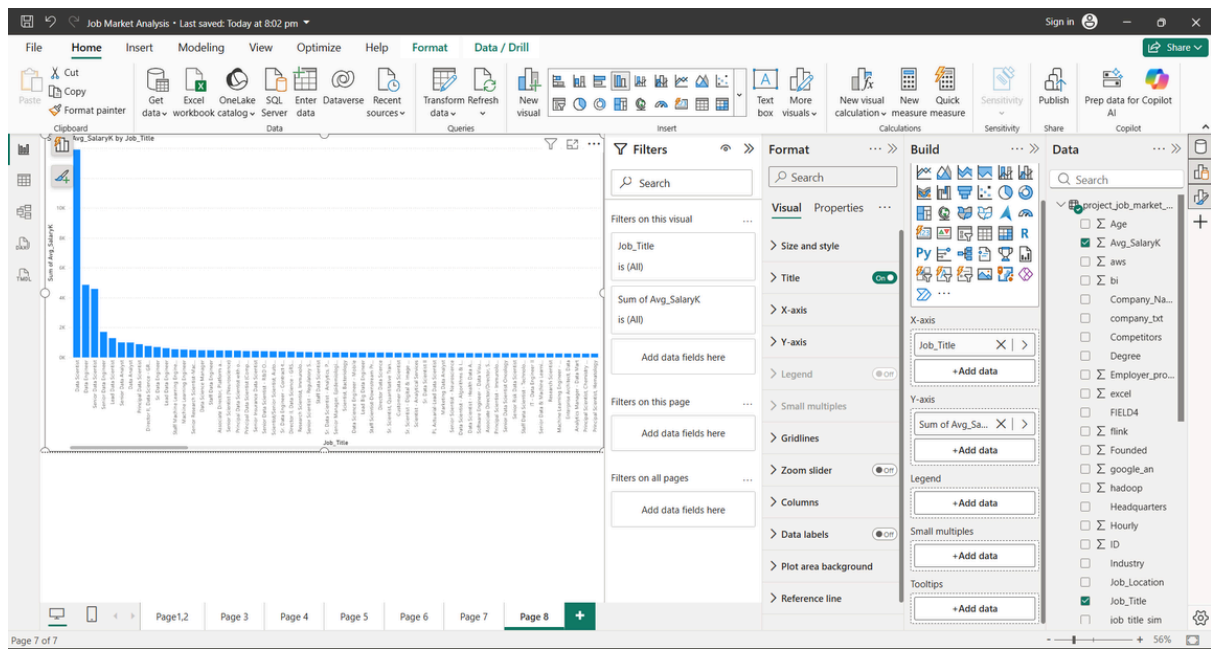
C. Average Salary by State



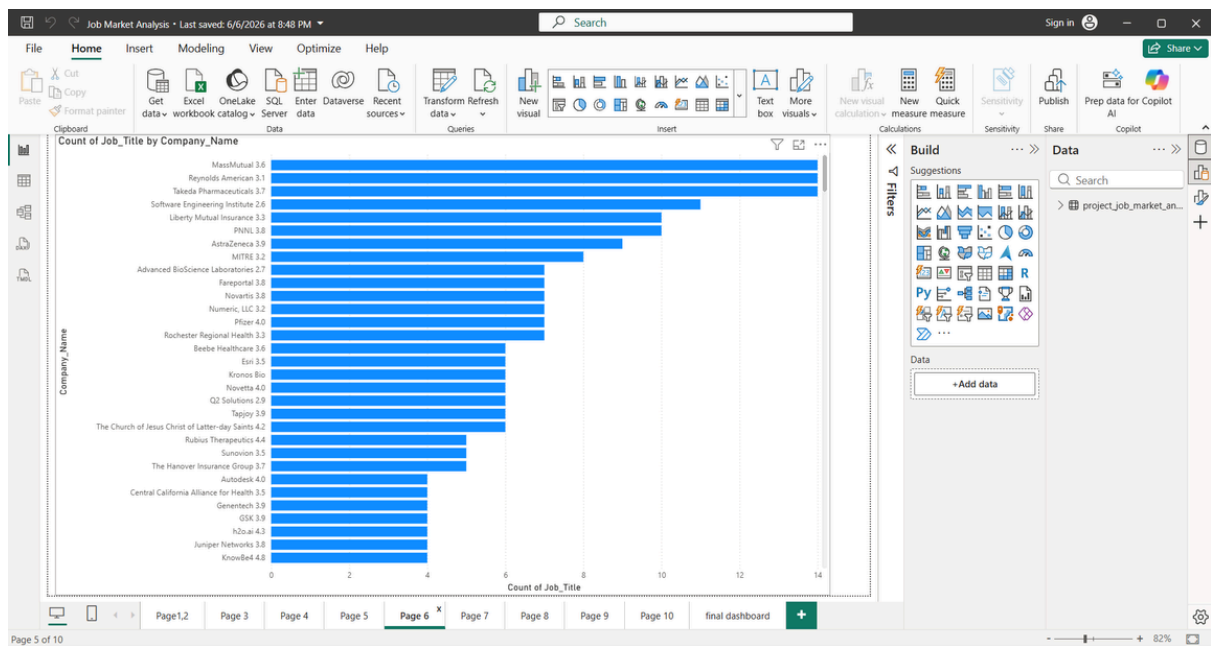
D. Top 5 Industries by Job Openings



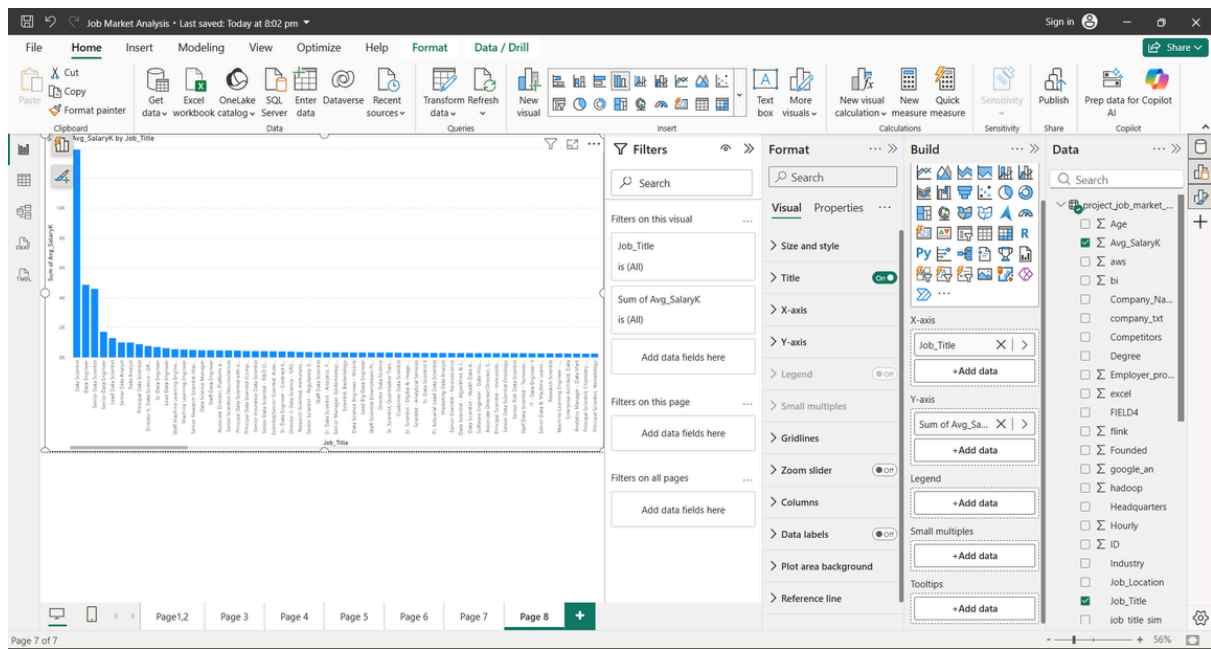
E. Companies with Maximum Job Openings



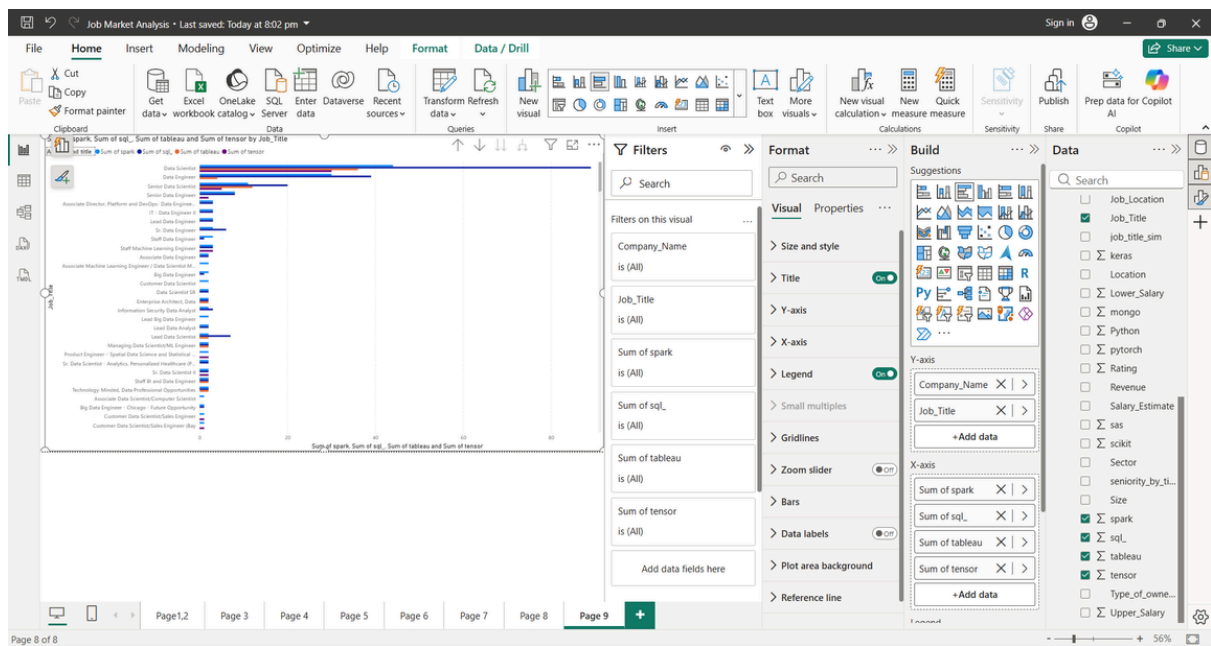
F. Job Titles with Most Jobs



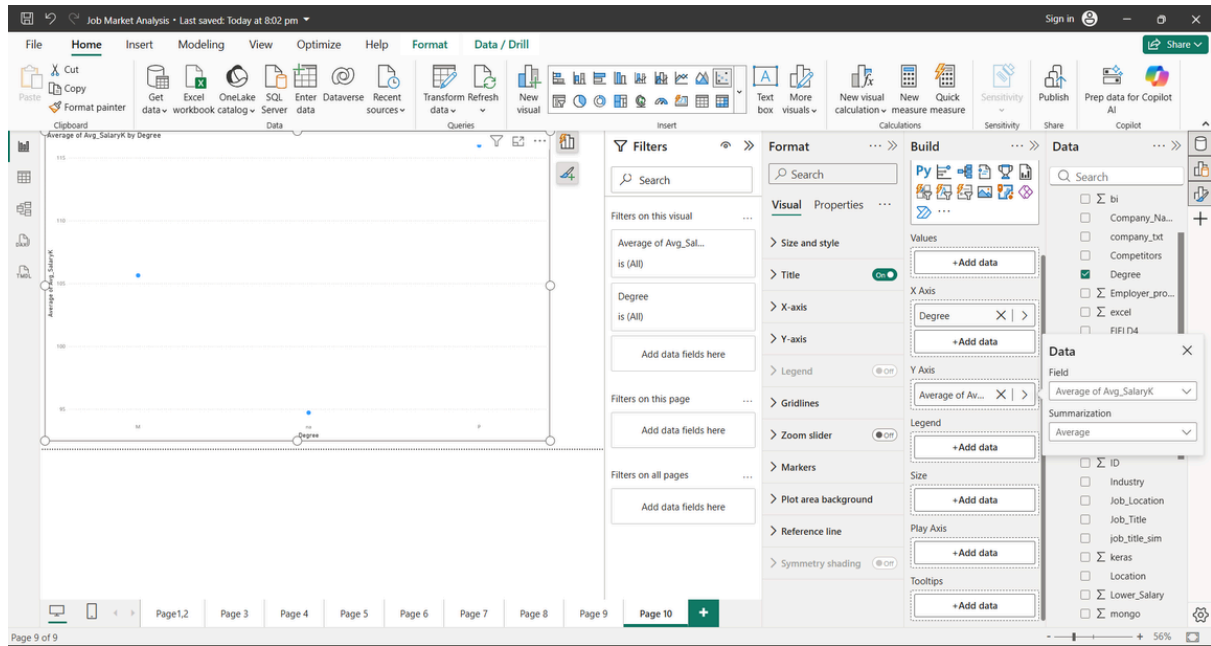
G. Salary by Job Title



H. Skills Analysis



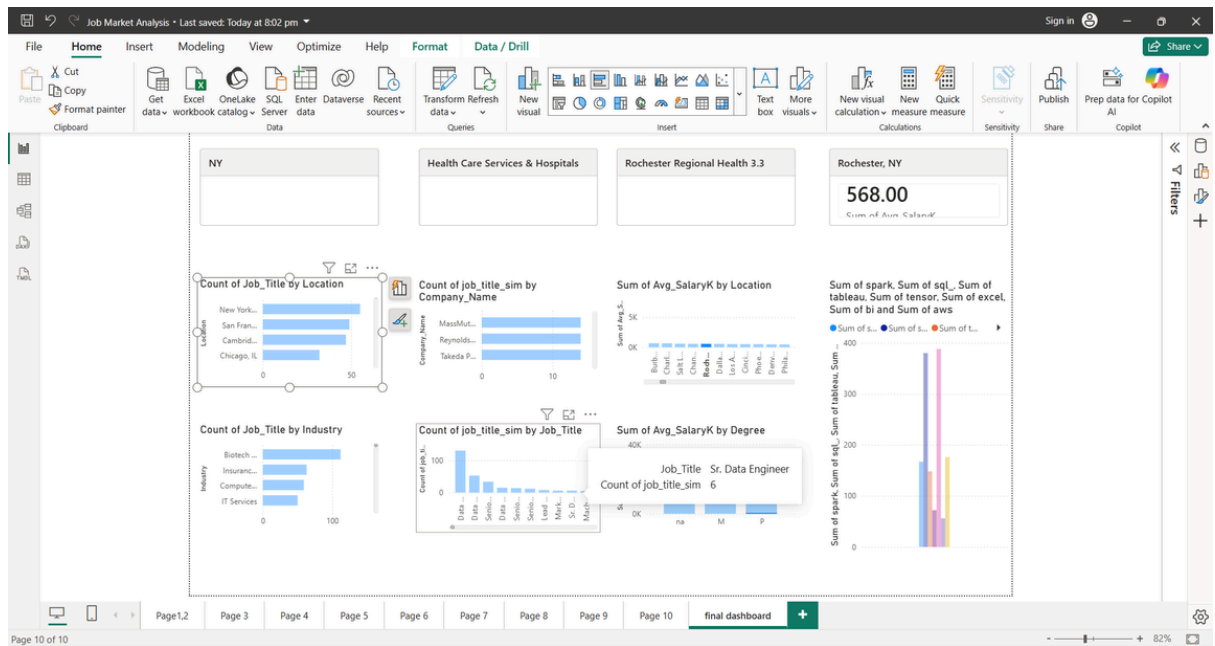
I. Degree vs Salary Analysis



Dashboard Development

The dashboard contains:

- KPI Cards (Total Jobs, Companies, Average Salary, Locations)
- State Analysis
- Industry Analysis
- Company Analysis
- Job Title Analysis
- Skills Analysis
- Education Analysis
- Interactive Slicers



Key Insights

1. Certain states contribute the highest share of job postings.2. Technology-related industries dominate hiring.3. A few companies account for a significant portion of openings.4. Data Scientist and Data Analyst are highly demanded roles.5. Advanced technical roles receive higher salaries.6. SQL and Python remain essential skills.7. Cloud and machine learning skills are increasingly valuable.8. Higher educational qualifications generally correspond to higher salaries.9. Salary ranges vary significantly across locations.10. Organizations prefer candidates with multiple technical skills.

Recommendations

1. Focus recruitment in high-demand regions.2. Prioritize SQL, Python, Spark, Tableau, and cloud skills.3. Invest in employee upskilling programs.4. Offer competitive salary packages.5. Strengthen industry-academia collaboration.6. Encourage certification and advanced training.7. Use analytics-driven hiring strategies.

Conclusion

The project successfully analyzed job market trends using Power BI. The dashboard provides valuable insights into hiring demand, salary structures, educational requirements, and skill expectations. The findings can support strategic workforce planning and career development.

References

- Power BI Documentation
- Course Materials
- Dataset Provided for Academic Project